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Doc. 230 – Hilbert space translation

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Preamble

This document formulates the **complete translatability** between the FFGFT mode formalism and standard Hilbert-space quantum mechanics. The translation is not an approximation or specialisation, but a bijective structural statement: every QM calculation can be mapped into FFGFT language, and conversely every FFGFT statement about quantum systems is reproducible in Hilbert-space language.

This is methodologically important: it means that FFGFT does not *replace* quantum mechanics, but *extends* it. Both formalisms yield the same measurable predictions (up to a ξ correction $\sim 10^{-5}$, currently below NISQ noise); they differ, however, in the ontological reading of several primitives – the Born rule, non-locality, the status of $\hbar$, the spacetime stage, singularities. This distinction is developed in the section “Operational equivalence and interpretive divergence”. Practitioners familiar with Hilbert-space language can use FFGFT without changing their tools – the bridge is the translation table below. (See also Doc. 202 §22 for the broader methodological placement.)

Notation

Geometric spaces

Symbol	Meaning
$\mathbb{R}$	real numbers, linear axis
$\mathbb{C}$	complex numbers
$\mathbb{C}^2$	two-dimensional complex vector space (standard qubit Hilbert space)
$\mathbb{Z}$	integers (for discrete windings)
S^1	circle, one compact loop
S^2	2-sphere, Bloch sphere of standard QM
$T^n = (S^1)^n$	n -torus, n compact loops
T^3	3-torus, three spatial windings (FFGFT mode space)
$T^4 = T^3 \times S^1$	4-torus, full FFGFT geometry incl. time winding
$\mathbb{R} \times S^1$	cylinder, limit of 2-torus with large major radius
$L^2(M)$	space of square-integrable functions on the manifold M
$\mathcal{H}$	Hilbert space (general)

FFGFT geometry parameters

Symbol	Meaning
$\xi_0 = 4/30000$	dimensionless FFGFT geometry parameter
ξ	alternative notation for ξ_0
$D_f = 2.999867$	fractal spacetime dimension
K_{frak}	fractal correction, $K_{\text{frak}} = 1 - 100\xi_0 \approx 0.9867$
$L_0 = \xi_0 \cdot L_P$	minimal FFGFT length scale (sub-Planck)
$L_\xi \approx 100 \mu\text{m}$	emergent vacuum length scale
v	Higgs vacuum expectation value, $\approx 246 \text{ GeV}$
E_0	FFGFT energy scale
m_h	Higgs mass, $\approx 125 \text{ GeV}$
λ_h	Higgs self-coupling
E_ξ	FFGFT energy scale for low-energy limit

Qubit coordinates and states

Symbol	Meaning
$ \psi\rangle$	quantum state (Hilbert-space vector)
$ 0\rangle, 1\rangle$	computational basis states
α, β	complex amplitudes, $ \alpha ^2 + \beta ^2 = 1$
z	FFGFT projection on computational axis, $z \in [-1, +1]$
r	FFGFT radial amplitude, $r \in [0, 1]$, $z^2 + r^2 = 1$
θ	FFGFT phase, $\theta \in [0, 2\pi)$
$\varphi_{n,l,j}$	FFGFT mode function on T^4
(n, l, j)	mode quantum numbers
(k_x, k_y, k_z, k_t)	wave vector components on T^4

Operators

Symbol	Meaning
$\sigma_x, \sigma_y, \sigma_z$	Pauli matrices
$\mathbb{1}$	identity matrix
U	unitary operator (quantum gate)
H	Hadamard gate (in gate tables); Hamiltonian (in Schrödinger equation)
T	T phase gate
CNOT	controlled-NOT gate (two-qubit)
$R_{\hat{n}}(\phi)$	Bloch-sphere rotation about axis $\hat{n}$ by angle ϕ
$\hat{n} \cdot \vec{\sigma}$	$n_x\sigma_x + n_y\sigma_y + n_z\sigma_z$
∇^2	Laplace operator

Standard Model quantities

Symbol	Meaning
α	fine-structure constant, $\alpha \approx 1/137.036$
m_i	mass of fermion i ($i = e, \mu, \tau, u, d, c, s, t, b$)
r_i	rational prefactor in $m_i = r_i \xi^{p_i} v$
p_i	winding exponent in $m_i = r_i \xi^{p_i} v$
$\hbar$	reduced Planck constant
c	speed of light

Other symbols

Symbol	Meaning
$\otimes$	tensor product
$\otimes_i$	tensor product over index i
$\oplus_n$	direct sum over index n
$ x\rangle$	basis state labelled x
$\langle\psi \phi\rangle$	scalar product
$ x $	absolute value of x
$\arg(x)$	argument (phase) of complex number x
δ_{ij}	Kronecker delta
ϵ_{ijk}	Levi-Civita tensor

Where we are, geometrically

Before the formal translation tables, a brief narrative picture of the underlying geometry. This matters because we are about to work with *cylindrical coordinates* – and the cylinder is not the full FFGFT geometry, but a convenient reduction.

The full FFGFT geometry is four-dimensional. The fundamental space on which all FFGFT statements live is the 4-torus $T^4 = T^3 \times S^1$ – three spatial winding directions and one temporal winding (Doc. 202 §13, Doc. 210). The wave function of a mode $\varphi_{n,l,j}$ is a

function on this 4D torus. From this geometry follow all masses, the fine-structure constant, cosmology – everything that distinguishes FFGFT from pure quantum mechanics.

For a single qubit, less suffices. When we want to describe just a single qubit – a two-level system at fixed time, without mass dynamics – most of these winding directions are not active. Two rotational degrees of freedom remain: one encoding “where between $|0\rangle$ and $|1\rangle$ we are” (this becomes our z), and one for the phase (this becomes our θ). The natural space for these two rotational degrees of freedom is a 2-torus $T^2 = S^1 \times S^1$.

From 2-torus to cylinder. If we now assume that one major radius R of the 2-torus is very large compared to the tube radius r , then the local geometry is no longer curved – it looks like a cylinder. The cylinder $\mathbb{R} \times S^1$ is thus the limit $R \rightarrow \infty$ of the 2-torus. Topologically one loop is “cut open”: from two circles we get a line plus a circle. In practice this means: in cylindrical coordinates (z, r, θ) , z is a *linear* axis (from -1 to $+1$), no longer cyclic.

From cylinder to Bloch sphere. If we contract the cylinder top and bottom to a point – the two “poles” – we obtain a sphere S^2 . This is the famous **Bloch sphere** of standard quantum mechanics. It is the second, further reduction: cylinder with merged ends.

Hierarchy of reductions.

Geometry	Dim.	Use
4-torus T^4	4	full FFGFT (masses, α , cosmology)
3-torus T^3	3	mode space at fixed time
2-torus T^2	2	single qubit, without reduction
Cylinder $\mathbb{R} \times S^1$	2	limit $R \rightarrow \infty$, local approximation
Bloch sphere S^2	2	poles contracted, standard QM

What the reductions cost. Each step of reduction loses something. From the 4-torus to the 3-torus one loses the time winding – and with it all mass statements (Doc. 210). From the 3-torus to the 2-torus one loses a spatial winding – and with it the connection to the third lepton generation and many spin structures. From the 2-torus to the cylinder one loses a *compact loop* – this is the weakest reduction, and exactly here sits the **fractal correction factor** $K_{\text{frak}} \approx 0.9867$ that appears in the bit-flip rotations of Doc. 175. K_{frak} is the “memory” of the cylinder of its toroidal origin.

What we translate here. The translation tables below apply to the *qubit sector*, i.e. the lower three rows of the hierarchy – 2-torus, cylinder, Bloch sphere. Standard quantum mechanics works on the Bloch sphere; FFGFT works on the cylinder (or the 2-torus, if one takes K_{frak} seriously). The translation between these levels is lossless, provided one carries the small geometric corrections (K_{frak} , ΔCHSH) along.

The full FFGFT geometry – the 4-torus – lies one step higher and provides what Hilbert-space QM cannot provide: the values of the input parameters (masses, α , ξ). This is discussed in §10 below.

.1 States: (z, r, θ) vs. (α, β)

Hilbert-space representation

In standard QM a qubit is a vector in $\mathbb{C}^2$:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1, \quad \alpha, \beta \in \mathbb{C}. \quad (1)$$

With global phase freedom this is a point on the Bloch sphere (Hopf fibration $S^3 \rightarrow S^2$).

FFGFT representation

In FFGFT the same qubit is a point (z, r, θ) on the cylinder/torus:

$$z \in [-1, 1], \quad r \in [0, 1], \quad \theta \in [0, 2\pi), \quad z^2 + r^2 = 1. \quad (2)$$

z is the projection onto the computational basis axis, r the radial superposition amplitude, θ the phase (Doc. 147 §1.2).

Bijjective translation

The two representations are linked by:

$$\alpha = \sqrt{\frac{1+z}{2}} \cdot e^{i\theta/2}, \quad \beta = \sqrt{\frac{1-z}{2}} \cdot e^{-i\theta/2}. \quad (3)$$

Conversely:

$$z = |\alpha|^2 - |\beta|^2, \quad r = 2|\alpha\beta|, \quad \theta = \arg(\alpha) - \arg(\beta). \quad (4)$$

Consistency checks:

- $|\alpha|^2 + |\beta|^2 = \frac{1+z}{2} + \frac{1-z}{2} = 1. \checkmark$
- $z^2 + r^2 = (|\alpha|^2 - |\beta|^2)^2 + 4|\alpha|^2|\beta|^2 = 1. \checkmark$
- $|0\rangle$ corresponds to $(z, r, \theta) = (1, 0, *)$. $\checkmark$
- $|1\rangle$ corresponds to $(z, r, \theta) = (-1, 0, *)$. $\checkmark$

Conceptual difference: in QM, $|\alpha|^2$ is a *probability* (Born rule). In FFGFT, $r^2 = 1 - z^2$ is an *energy density* of the radial configuration. The two quantities are *numerically equal* at large statistics (Doc. 147), but conceptually different: FFGFT is a deterministic geometric model whose statistical predictions coincide with the Born rule.

.2 Operators: Pauli matrices vs. rotations

Hilbert-space operators

The three Pauli matrices generate all single-qubit operations:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (5)$$

Algebra: $\sigma_i \sigma_j = \delta_{ij} \mathbb{1} + i\epsilon_{ijk} \sigma_k$.

FFGFT rotations

On the Bloch cylinder, the three Pauli matrices correspond to three geometric rotations (Doc. 175 §3):

- σ_z – rotation about the cylinder axis (phase rotation $\theta \rightarrow \theta + \pi$).
- σ_x – 2D rotation in the (z, r) plane by angle π (bit flip).
- σ_y – orthogonal 2D rotation, combining σ_x and phase rotation.

Translation table

Operator	Hilbert-space matrix	FFGFT geometry
σ_z	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$	Phase rotation $\theta \rightarrow \theta + \pi$
σ_x	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$	(z, r) -rotation by πK_{frak}
σ_y	$\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$	Rotation with phase tilt
H	$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$	Swap $z \leftrightarrow r$, $\theta \rightarrow \theta + \pi/2$
T	$\begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$	Phase rotation $\theta \rightarrow \theta + \pi/4$

The only FFGFT-specific deviation is the factor $K_{\text{frak}} \approx 0.9867$ in the σ_x rotation (Doc. 175). This deviation follows from the fractal spacetime dimension $D_f = 2.999867$ and is a *testable prediction*: all π -gates deviate systematically by $\Delta\alpha \approx 0.013\%$ from the ideal value. In Hilbert-space formalism this would be an ad hoc correction modeled as hardware noise; in FFGFT it follows from the geometry.

.3 Quantum gates: complete translatability

Single-qubit gates

Every unitary single-qubit gate $U \in U(2)$ can be represented as a Bloch-sphere rotation:

$$U = e^{i\alpha} R_{\hat{n}}(\phi) = e^{i\alpha} (\cos(\phi/2) \mathbb{1} - i \sin(\phi/2) \hat{n} \cdot \vec{\sigma}). \quad (6)$$

In FFGFT this is a rotation of the (z, r, θ) point about axis $\hat{n}$ by angle ϕ . The translation is exact: $U|\psi_{\text{QM}}\rangle$ corresponds to the rotation of the FFGFT cylinder point about $\hat{n}$ by ϕ .

Two-qubit gates

The CNOT gate:

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \quad (7)$$

translates in FFGFT as: *if $z_A = -1$ (control qubit on $|1\rangle$), perform a bit flip on qubit B*. This is geometric conditioning: the rotation of one cylinder point depends on the z value of the other. Topologically this corresponds to a linkage of the two cylinders (entanglement).

Universality

The set $\{H, T, \text{CNOT}\}$ is universal for QM (Solovay-Kitaev theorem). Since each of these gates is representable as a geometric operation in FFGFT, FFGFT is *fully universal for quantum computation*. There is no QM statement that cannot be formulated in FFGFT.

.4 Tensor products and many-qubit spaces

Hilbert-space construction

For n qubits the Hilbert space is $\mathcal{H}_n = \bigotimes_{i=1}^n \mathbb{C}^2$ with dimension 2^n . A general state is:

$$|\Psi\rangle = \sum_{x \in \{0,1\}^n} \alpha_x |x\rangle, \quad \sum_x |\alpha_x|^2 = 1. \quad (8)$$

FFGFT product space

In FFGFT n qubits correspond to n independent mode configurations (z_i, r_i, θ_i) on a shared torus geometry. Entanglement is represented as a *topological linkage*: entangled qubits are nodes of the mode configuration that cannot be reduced to product form (Doc. 175 §4).

Bell states

The maximally entangled Bell state

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \quad (9)$$

has in FFGFT a geometric characterisation as a *topologically linked cylinder configuration*, in which the individual (z_i, r_i, θ_i) values are not independent.

CHSH prediction: standard QM gives $|\langle \text{CHSH} \rangle| \leq 2\sqrt{2}$ (Tsirelson bound). FFGFT predicts a small, geometrically motivated deviation:

$$\Delta \text{CHSH} \approx 10^{-5} \quad (\text{geometric, from } \xi \text{ and } K_{\text{frak}}). \quad (10)$$

This deviation lies below the current hardware noise floor and is an open experimental task for high-precision Bell tests (Doc. 175 §6, Doc. 023).

Terminology note: $\Delta \text{CHSH} \approx 10^{-5}$ is the *elementary* deviation per measurement, derived from $\xi/(2\pi)$. This is distinct from the *cumulative* value at $N = 73$ measurements in Doc. 022 ($\approx 5 \times 10^{-4}$). The two quantities are not directly comparable; 10^{-5} is not reachable at any finite N from the cumulative formula in Doc. 022.

Exponential growth

Both formalisms reproduce the exponential growth $\dim = 2^n$. In Hilbert space through tensor product; in FFGFT through n independent mode configurations with topological entanglements. The number of real parameters per state is in both cases $2 \cdot 2^n - 2$.

.5 Measurement

Hilbert-space Born rule

A measurement in the computational basis yields outcome 0 with probability $|\alpha|^2$ and outcome 1 with probability $|\beta|^2$. The state collapses to $|0\rangle$ or $|1\rangle$ respectively.

FFGFT measurement as polar transition

In FFGFT, measurement is *not* a probabilistic collapse, but a *geometric interaction* that drives the (z, r, θ) point to the nearest stable extremum ($z = +1$ or $z = -1$). This is a deterministic motion in mode space – but due to the unknown phase θ and tiny variations in initial conditions, it is *statistically* indistinguishable from Born-rule behaviour.

Numerical equivalence

At large statistics (many repetitions), the FFGFT measurement distribution converges to the Born rule:

$$P_{\text{FFGFT}}(z = +1) = \frac{1+z}{2} = |\alpha|^2 = P_{\text{QM}}(0). \quad (11)$$

The two formalisms yield *identical* statistical predictions.

Conceptual difference: the Bell-test realism debate resolves geometrically in FFGFT, since entanglement is represented as a topological link – not as nonlocal correlation. The experimental predictions are the same.

.6 Unitary evolution: Schrödinger equation

Hilbert-space Schrödinger

The time evolution in QM is:

$$i\hbar \frac{\partial |\psi\rangle}{\partial t} = H|\psi\rangle. \quad (12)$$

FFGFT mode evolution

In FFGFT the time evolution is the deterministic motion of the mode configuration $\varphi_{n,l,j}$ on the torus $T^3 \times \mathbb{R}$ (Doc. 202 §13). The corresponding equation of motion – the FFGFT mode equation – reduces in the low-energy limit ($E \ll E_\xi$) to the Schrödinger equation:

$$i\hbar \frac{\partial \varphi_{n,l,j}}{\partial t} \xrightarrow{E \ll E_\xi} -\frac{\hbar^2}{2m} \nabla^2 \varphi_{n,l,j} + V \varphi_{n,l,j}. \quad (13)$$

At higher energies, ξ corrections appear, which in the Hilbert-space formalism manifest as Hamiltonian modifications.

.7 What is fully translatable

All Hilbert-space statements map into FFGFT:

- States, operators, gates (with or without K_{frak} correction),
- Tensor products, entanglement, Bell states, GHZ states,
- Measurement (statistical predictions identical to Born rule),
- Unitary evolution (Schrödinger in low-energy limit),
- Algorithms (Shor, Grover, VQE, etc.).

All FFGFT QM statements map into Hilbert space:

- The (z, r, θ) configuration corresponds uniquely to the (α, β) vector (bridge (3)),
- Geometric rotations correspond to unitary operators,
- Topological entanglements correspond to entangled Hilbert-space vectors.

Consequence: every computation that can be performed in one formalism can be performed in the other – with identical end result. Practitioners can choose the formalism familiar to them.

.8 What is FFGFT-specific and not representable in pure Hilbert space

The translatability holds for *statements about quantum systems*. It does not hold for *statements about the input values* of quantum mechanics:

- The geometric parameter $\xi_0 = 4/30000$ appears in the Hilbert-space formalism only as an external input. In FFGFT it follows from Higgs matching $\xi_0 = \lambda_h^2 v^2 / (16\pi^3 m_h^2)$ (Doc. 006).
- The fermion masses $m_e, m_\mu, m_\tau, m_u, \dots$ appear in Hilbert space as external Yukawa couplings. In FFGFT they all follow parameter-free from $m_i = r_i \xi_0^{p_i} v$ (Doc. 006).
- The fine-structure constant $\alpha = \xi_0 \cdot E_0^2$ is derivable in FFGFT, an input in Hilbert space.
- The fractal correction factor $K_{\text{frak}} = 1 - 100\xi_0 \approx 0.9867$ appears in FFGFT gates as a geometric deviation. In Hilbert space it would have to be modeled as hardware noise.

This is the structural asymmetry:

- Hilbert space: has all tools for QM calculations, but no explanation for the input values (masses, couplings, α).
- FFGFT: provides both the tools for QM calculations *and* the input values from a single geometry.

In this sense QM is a subset of FFGFT (see Doc. 202 §22): QM is FFGFT, restricted to the question *how systems behave*, without the question *why the constants take these values*.

.9 Consequence: extension, not replacement

FFGFT does not replace quantum mechanics. It extends it to a level where the input values themselves follow from geometry.

For practitioners this means:

- Those who calculate with Hilbert-space tools can continue to do so. The translation (3) and the operator table allow switching to the FFGFT formalism at any time, without loss.
- Those who calculate with the FFGFT mode formulation additionally have access to masses, couplings, and α from the same geometry.
- Both languages are *interoperable*. This is precisely the “translatability without contradiction” from Doc. 202 §22: other frameworks can map FFGFT statements into their language, provided they use the translation table.

Methodological clarification: FFGFT does not claim to be the only correct representation of quantum mechanics. It is *one* representation that, in addition to the standard QM statements, provides the values of the fundamental constants. Those who do not need this added value can ignore FFGFT – the standard QM statements remain valid.

.10 Operational equivalence and interpretive divergence

The full translatability shown in § 3ff. must not be read as ontological identity. *Operationally*, FFGFT and standard quantum mechanics agree on every measurable outcome up to ξ corrections – this is the content of the preceding sections. *Interpretively*, they

diverge at several points where the standard interpretation makes claims that FFGFT reads as artefacts of the chosen carrier assumptions, not as ontological properties of nature. This distinction matters, because otherwise the limit $\xi \rightarrow 0$ can be misread as "FFGFT is just standard QM with a small correction". It reproduces the *numbers* of standard QM, not necessarily its ontological reading.

Five concrete points where the readings part company:

- **Born rule as irreducible randomness.** Standard reading: $|\psi|^2$ as ontological probability; the measurement outcome is fundamentally random. FFGFT reading: $|\psi|^2$ is the correct distribution of *observations*, but it follows from a deterministic polar-transition dynamics on T^4 (§ Measurement). Probability arises at the *measurement apparatus*, not in the natural process itself. Operationally identical; interpretively a different location of stochasticity.
- **Non-locality as an ontological property.** Standard reading of Bell correlations: spooky action at a distance or genuine non-locality in $\mathbb{R}^3$ space. FFGFT reading: entanglement is a topological connection on T^4 – geometric proximity in a closed geometry, not action across an open space (§ Bell states and Doc. 193 [?]). The *experimental predictions* are identical; the *ontological reading* differs structurally.
- **$\hbar$ as an ontological primitive.** Standard reading: $\hbar$ is a fundamental constant of nature. FFGFT reading: $\hbar$ is an SI conversion factor between energy and time measures, following from the closure condition $S = h$ on T^4 (Doc. 001a, Doc. 260 § 2). Numerically identical; ontologically not primitive but derived. This has consequences for interpretations such as "the Planck scale as a fundamental limit of describability", which FFGFT reads differently.
- **Spacetime as fundamental stage.** Standard reading in general relativity: spacetime is the given manifold on which fields live. FFGFT reading: spacetime is *not* presupposed but derived from the T^4 identification geometry (Doc. 260 §1, in common with UC: "derives without presupposing spacetime"). Questions such as "what was before the Big Bang" are, in FFGFT, posed as artefacts of the $\mathbb{R}^3 + t$ assumption; on T^4 they have no geometric substance.

- **Singularities as physical realities.** Standard reading: black hole, Big Bang as genuine singularities of the manifold. FFGFT reading: because T^4 is closed through identifications, these singularities appear structurally differently – as endpoints of the carrier model, not as endpoints of physics. Operationally the predictions outside the singular region coincide; *inside*, the reading diverges.

Methodological status. The five points are not empirical claims against standard QM or general relativity – the predictions agree, up to ξ . They are *interpretive differences* that become visible once the question turns to the *ontological carrier* of the theory. Anyone who only computes with the numbers can ignore them. Anyone who wants to know *why* the numbers are what they are – the values of the fundamental constants, the Bell correlations, the periodicity of the measurement spectrum – will find that the five points become testable structural statements.

Operational equivalence. Within the reach of present measurement precision (NISQ level $\sim 10^{-2}$, optical precision experiments $\sim 10^{-12}$ for Bell tests), FFGFT and standard QM deliver the same predictions. The ξ correction ($\sim 10^{-5}$) is not resolvable on current hardware (empirically checked, hardware validation May 2026 on IBM Heron r2, Doc. 147 §8). Operationally, therefore: equivalent descriptions, one with and one without an ontological carrier.

.11 Summary

What is fully translatable

The translation table FFGFT $\leftrightarrow$ Hilbert-space QM is complete at the formal level:

1. **States:** bijective bridge $\alpha = \sqrt{(1+z)/2} e^{i\theta/2}$, $\beta = \sqrt{(1-z)/2} e^{-i\theta/2}$.
2. **Operators:** Pauli matrices $\leftrightarrow$ Bloch rotations, with FFGFT correction K_{frac} in σ_x .
3. **Gates:** all standard gates (H , T , CNOT, ...) in FFGFT as geometric transformations.
4. **Tensor products:** 2^n growth reproduced, entanglement as topological linkage.

5. **Measurement:** statistically identical to Born rule, conceptually deterministic geometry.
6. **Evolution:** Schrödinger equation as low-energy limit of FFGFT mode evolution.

What FFGFT additionally provides

Within the FFGFT formalism, but beyond what pure Hilbert-space QM delivers:

- parameter-free derivation of α , m_i , ξ from a single geometry,
- geometric explanation of the fractal correction factor K_{frak} in π -gates (testable in high-precision quantum gate experiments),
- geometric resolution of the Bell-test realism debate (entanglement as topology, not as non-locality),
- small, geometrically motivated CHSH deviation $\sim 10^{-5}$ as a testable prediction.

Where the interpretation parts company

The translatability is *operational*, not ontological. Standard QM and FFGFT share the measurable predictions but diverge at five points in their reading (details in the section "Operational equivalence and interpretive divergence"):

- Born rule as irreducible randomness vs. deterministic polar-transition dynamics on T^4 ,
- non-locality in $\mathbb{R}^3$ vs. topological connection on T^4 ,
- $\hbar$ as ontological primitive vs. SI conversion factor from $S = h$,
- spacetime as given stage vs. derived from T^4 geometry,
- singularities as physical endpoints vs. endpoints of the carrier model.

Operationally identical; interpretively distinct. Anyone who only computes with the numbers can ignore the difference – the predictions are the same. Anyone who asks about the ontological carrier finds five structurally testable points.

Methodological position

FFGFT confirms quantum mechanics at the predictive level and extends it at the structural level. It refutes nothing that QM says *measurably*; it does, however, propose a different ontological reading of several primitives. The translatability $\leftrightarrow$ Hilbert space is, in this sense, a *consistency proof*: any internally consistent theory should be expressible in multiple mathematical languages, and the existence of such translations is evidence of structural soundness – regardless of which of the languages one regards as ontologically primary.